Linear Algebra midterm 2019 Fall *Name* :

1. Calculate the results of the following matrix operations.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \quad , \; B = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} \quad , \; C = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad , \; D = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 2 & 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & 5 \\ 14 & 32 \end{bmatrix} \qquad D = \begin{bmatrix} 30 \end{bmatrix}$$

2. Find the matrix determinants. (**Hint:** The determinant of a triangular matrix is equal to the product of the elements in its diagonal.)

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & a & b & c \\ 0 & 2 & d & e \\ 0 & 0 & 3 & f \\ 0 & 0 & 0 & 4 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 2 & 0 & 0 \\ b & c & 3 & 0 \\ d & e & f & 4 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & c & 1 & 0 \\ d & e & f & 1 \end{bmatrix}.$$

$$|A| = 1 \cdot 2 \cdot 3 \cdot 4 \qquad |B| = 1 \cdot 2 \cdot 3 \cdot 4 \qquad |C| = 1 \cdot 2 \cdot 3 \cdot 4 \qquad |D| = 1$$

3. Find the matrix determinants. (**Hint:** The determinant of a triangular matrix is equal to the product of the elements in its diagonal.)

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 \\ 2 & 1 & 3 & 4 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 2 & 1 & 0 & 3 \\ 2 & 1 & 4 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}.$$

triangular matrix: $|A| = 1 \cdot 2 \cdot 3 \cdot 4 = 24$

single column swap: $|B| = (-1)24 = -24$

double column swap: $|C| = (-1)^2 \; 24 = 24$

single row swap: $|D| = (-1)24 = -24$

4. Which ones of the following matrices CERTAINLY have a zero determinant. EXPLAIN!

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ -1 & 1 & -1 & 1 & -1 & 1 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ -2 & -2 & -2 & -2 & -2 & -2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 8 & 10 & 12 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}.$$

$|A| = 0$ since it has zero row.

$|B| = 0$ since 3rd row = -(1st row).

$|C| = 0$ since 3rd row = 2(1st row)+2(2nd row).

$|D| = 0$ since 3rd row = 2(2nd row).

5. Let k be an integer. Let A, B, and C be $n \times n$ matrices with the determinants $|A|$, $|B|$, and $|C|$. Find the following matrix determinants.

$|A \cdot B| = function(|A| \; , \; |B|) =$ $|A| \cdot |B|$

$|A \cdot B \cdot C| = function(|A| \; , \; |B| \; , \; |C|) =$ $|A| \cdot |B| \cdot |C|$

$|k \cdot A| = function(k \; , \; |A|) =$ $k^n |A|$

$|A^T \cdot B \cdot C^T| = function(|A| \; , \; |B| \; , \; |C|) =$ $|A| \cdot |B| \cdot |C|$

6. Obtain the ECHELON form of the matrix

$$A = \begin{bmatrix} 2 & -2 & 4 & 2 \\ 4 & -3 & 4 & 3 \\ -2 & 3 & 1 & 1 \\ 6 & -3 & 9 & 7 \end{bmatrix}$$

$$A' = \begin{bmatrix} 2 & -2 & 4 & 2 \\ 0 & 1 & -4 & -1 \\ 0 & 1 & 5 & 3 \\ 0 & 3 & -3 & 1 \end{bmatrix}$$

$$A'' = \begin{bmatrix} 2 & -2 & 4 & 2 \\ 0 & 1 & -4 & -1 \\ 0 & 0 & 9 & 4 \\ 0 & 0 & 9 & 4 \end{bmatrix}$$

$$A''' = \begin{bmatrix} 2 & -2 & 4 & 2 \\ 0 & 1 & -4 & -1 \\ 0 & 0 & 9 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

7. Obtain the REDUCED ECHELON form of the matrix

$$U = \begin{bmatrix} 2 & -6 & 4 & -4 & -2 \\ 0 & 3 & 3 & -4 & 1 \\ 0 & 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$U' = \begin{bmatrix} 2 & 0 & 10 & -12 & 0 \\ 0 & 3 & 3 & -4 & 1 \\ 0 & 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$U'' = \begin{bmatrix} 2 & 0 & 10 & 0 & 3 \\ 0 & 3 & 3 & 0 & 2 \\ 0 & 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$U''' = \begin{bmatrix} 1 & 0 & 5 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

8. Let the matrix A, its echelon form U, and its reduced echelon form R be as follows.

$$A=\begin{bmatrix} 2 & 3 & -2 & 4 & -1 \\ -2 & 1 & 4 & 1 & 15 \\ 4 & -2 & -5 & 1 & 3 \\ -6 & 3 & 3 & -1 & 6 \end{bmatrix} \xrightarrow{\text{Echelon}} U=\begin{bmatrix} 2 & 3 & -2 & 4 & -1 \\ 0 & 4 & 2 & 5 & 14 \\ 0 & 0 & 3 & 3 & 33 \\ 0 & 0 & 0 & 5 & 60 \end{bmatrix} \xrightarrow{\substack{\text{Reduced}\\ \text{Echelon}}} R=\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -11 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 12 \end{bmatrix}$$

a) Solve the matrix equation $A\mathbf{x}=\mathbf{0}$ where $\mathbf{x}=[x_1,\ x_2,\ x_3,\ x_4,\ x_5]^T$. **Hint:** $A\mathbf{x}=\mathbf{0} \Leftrightarrow U\mathbf{x}=\mathbf{0} \Leftrightarrow R\mathbf{x}=\mathbf{0}$.

$$\overbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -11 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 12 \end{bmatrix}}^{\mathbf{R}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 9x_5 \\ 11x_5 \\ x_5 \\ -12x_5 \\ x_5 \end{bmatrix} \qquad x_5 \text{ is free variable}$$

b) Solve the following linear quations using the above result.

$$\begin{aligned} 2x_1+3x_2-2x_3+4x_4&=-1 \\ -2x_1+x_2+4x_3+x_4&=15 \\ 4x_1-2x_2-5x_3+x_4&=3 \\ -6x_1+3x_2+3x_3-x_4&=6 \end{aligned} \Rightarrow \begin{aligned} 2x_1+3x_2-2x_3+4x_4-1(-1)&=0 \\ -2x_1+x_2+4x_3+x_4+15(-1)&=0 \\ 4x_1-2x_2-5x_3+x_4+3(-1)&=0 \\ -6x_1+3x_2+3x_3-x_4+6(-1)&=0 \end{aligned} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\Bigg|_{x_5=-1} = \begin{bmatrix} -9 \\ -11 \\ -1 \\ 12 \end{bmatrix}$$

9. Let the matrix A, its echelon form U, and its reduced echelon form R be as follows.

$$A=\begin{bmatrix} 1 & -2 & -2 & -3 & -3 \\ -2 & 4 & 5 & 8 & 7 \\ 1 & -2 & 0 & 1 & -1 \\ 2 & -4 & -3 & -4 & -5 \end{bmatrix} \xrightarrow{\text{Echelon}} U=\begin{bmatrix} 1 & -2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{\text{Reduced}\\ \text{Echelon}}} R=\begin{bmatrix} 1 & -2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

a) Solve the matrix equation $A\mathbf{x}=\mathbf{0}$ where $\mathbf{x}=[x_1,\ x_2,\ x_3,\ x_4,\ x_5]^T$. **Hint:** $A\mathbf{x}=\mathbf{0} \Leftrightarrow U\mathbf{x}=\mathbf{0} \Leftrightarrow R\mathbf{x}=\mathbf{0}$.

$$\overbrace{\begin{bmatrix} 1 & -2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}^{\mathbf{R}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_5-x_4+2x_2 \\ x_2 \\ -x_5-2x_4 \\ x_4 \\ x_5 \end{bmatrix}$$

b) Find the null space of A and its dimension. Find the matrix rank (Number of the linearly independent rows or columns).

$$\text{null}(A)=x_2\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}+x_4\begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}+x_5\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$\dim(\text{null}(A))=3$
since null space is spanned by 3 vectors
$\text{rank}(A)=2$
since there are 2 pivots in U and R. Also,
since $n-\dim(\text{null}(A))=5-3=2$

10. Let $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Find its minors, cofactor matrix, adjoint matrix, determinant, and its inverse.

$$A_{1,1}=4 \qquad A_{1,2}=3 \qquad A_{2,1}=2 \qquad A_{2,2}=1$$

$$\text{cofactor}(A)=\begin{bmatrix} |A_{1,1}| & -|A_{1,2}| \\ -|A_{2,1}| & |A_{2,2}| \end{bmatrix}=\begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$$

$$\text{adjoint}(A)=\begin{bmatrix} |A_{1,1}| & -|A_{2,1}| \\ -|A_{1,2}| & |A_{2,2}| \end{bmatrix}=\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$|A|=1\cdot 4-2\cdot 3=-2 \qquad , A^{-1}=\frac{\text{adjoint}(A)}{|A|}=\frac{1}{-2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}=\begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$